

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS
PAPER 1

4004/1
2 hours 30 minutes

NOVEMBER 2025 SESSION

Additional materials:
Mathematical Instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of each page. Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

If working is needed for any question, it must be shown in the space below that question. Omission of essential working will result in loss of marks.

Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. This paper is marked out of **100**.

This question paper consists of 21 printed pages.

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[Turn over



Answer **all** questions.

**NEITHER MATHEMATICAL TABLES NOR SLIDE RULES
NOR CALCULATORS MAY BE USED IN THIS PAPER.**

- 1 (a) Evaluate $-2 + 3 \times 4 - 4$.

Answer (a)

[1]

- (b) Simplify $5,7 \div 0,19$.

Answer (b)

[1]

- (c) Simplify $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$.

Answer (c)

[1]

- 2 (a) Estimate 4, 9985 correct to 3 significant figures.

Answer (a)

[1]



Candidate Name

Centre Number

Candidate Number

3

- (b) A line l measures 7cm correct to the nearest centimetre. Express the limits of the length of the line in the form $a \leq l < b$, where a and b are measurements.

Answer (b)

[2]

- 3 Simplify $43_5 + 101110_2$, giving the answer in base two.

Answer

[3]

- 4 Convert

- (a) $3,5 \text{ m}^2$ to cm^2 ,

Answer (a)

[1]

Candidate Name

Centre Number

Candidate Number

4

(b) $0,75\text{g/cm}^3$ to kg/m^3 .

Answer (b)

[2]

- 5 A man sitting on top of a cliff sees a boy 200m away from the bottom of the cliff.
The angle of depression of the boy from the man is 30° .

Use as much of the following information as is necessary

[Sin $30^\circ = 0,500$; Cos $30^\circ = 0,866$; Tan $30^\circ = 0,577$]

- (a) State the angle of elevation of the man from the boy.

Answer (a)

[1]

- (b) Calculate the height of the cliff.

Answer (b)

[2]



Candidate Name

Centre Number

Candidate Number

5

- 6 (a) Convert $N47^\circ W$ to a three - figure bearing.

Answer (a)

[1]

- (b) P and Q are points on level ground. The bearing of P from Q is $N47^\circ W$.
Find the three - figure bearing of Q from P.

Answer (b)

[2]

- 7 Given that $\log x = 0, 2$ and $\log y = 0, 3$
find

- (a) $\log(xy)$,

Answer (a)

[2]

Candidate Name

Centre Number

Centre Number

Candidate Number

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6

(b) $\log \left(\frac{1}{y} \right)$

Answer (b)

[2]

- 8 (a) Mrs Moyo took ZW\$34 000 to a bank to exchange for US dollars and obtained US\$ 400.

Find the exchange rate used, in the form US\$1 : ZW\$.

Answer (a)

[2]

- (b) She imported a machine worth US\$345 including 15% excise duty. Calculate how much she paid as excise duty.

Answer (b)

[2]



Candidate Name

Centre Number

Candidate Number

7

- 9 Triangle XYZ is right-angled at Y and $YZ = 12\text{cm}$.

The area of triangle XYZ is 30cm^2 .

Find the

- (a) length of XY,

Answer (a)

[2]

- (b) length of XZ.

Answer (b)

[2]

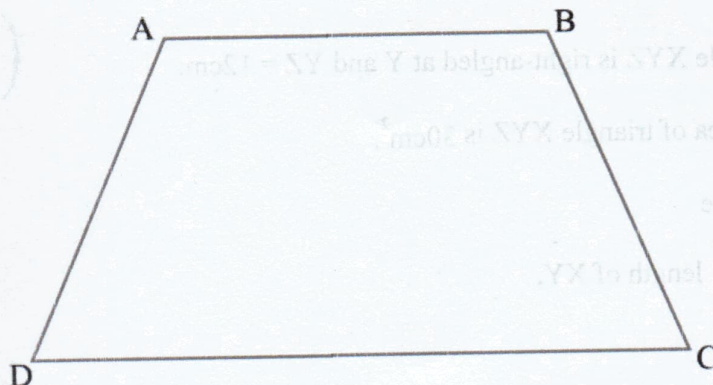
Candidate Name

Centre Number

Candidate Number

8

10



The diagram shows an isosceles trapezium ABCD.

(a) Name

(i) a pair of equal sides,

Answer (a)(i)

[1]

(ii) two pairs of equal angles.

Answer (b)(ii)

[2]

Candidate Name

Centre Number

Centre Number

Candidate Number

9

(b) State for the shape the

(i) number of lines of symmetry,

Answer (b)(i)

[1]

(ii) order of rotational symmetry.

Answer (b)(ii)

[1]

11 (a) Remove the brackets and simplify

$$2x + y - (x - 2y).$$

Answer (a)

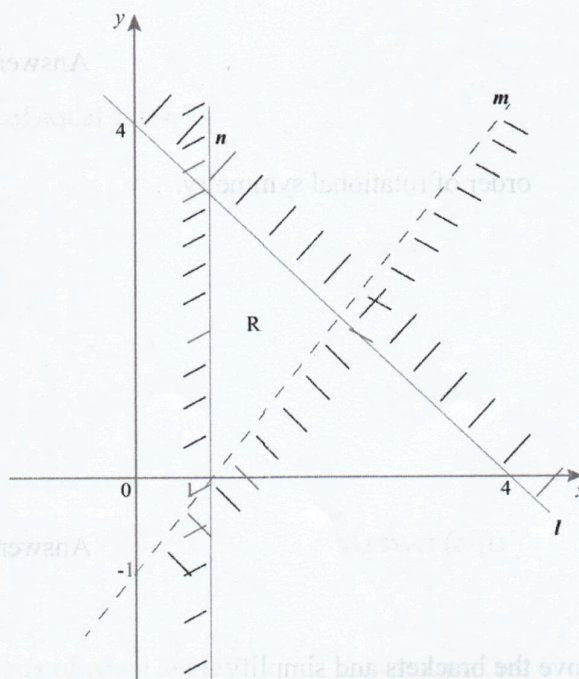
[2]

(b) Simplify $\frac{w^2 + w - 6}{w^2 - 9}$.

Answer (b)

[3]

12



Lines l , m and n are the boundaries of the unshaded region R which contains the solution set of three simultaneous inequalities.

Find the inequality whose boundary is line.

a) l ,

Answer (a)

[2]



Candidate Name

Centre Number

Candidate Number

11

(b)

m,

Answer (b)

[2]

(c)

n.

Answer (c)

[1]

13

(a)

Express the ratio 1,7:8,5 in the form 1:n.

Answer (a)

[2]

Candidate Name

Centre Number

Candidate Number

12

- (b) P, Q and R share \$36 so that for every \$1 that P gets, Q gets \$2 and for every \$1 that Q gets, R gets \$3. Find R's share.

Answer (b)

[3]

- 14 $A = \{x : 1 \leq x \leq 20, x \in \mathbb{Z} \text{ } x \text{ is a prime number}\},$
 $B = \{y : 0 < y \leq 20, y \in \mathbb{Z} \text{ } y \text{ is a perfect square number}\}.$

- (a) List the elements of

- (i) A,

Answer (a)(i)

[1]

- (ii) $A \cap B.$

Answer (a)(ii)

[2]



Candidate Name

Centre Number

Candidate Number

13

(b) Find the

(i) smallest value of $\frac{x}{y}$,

Answer (b)(i)

[1]

(ii) largest value of $x - y$.

Answer (b)(ii)

[1]

- 15 Jane noted the number of letters in each of the 25 words on a list. The results are given in the table.

Number of letters	2	3	4	5	6	7	8
Frequency	2	6	5	5	4	0	3

For this distribution,

(a) state the mode,

Answer (a)

[1]



Candidate Name	Centre Number	Candidate Number

14

(b) find the median,

Answer (b)

[1]

(c) calculate the mean.

Answer (c)

[3]

16 On a map, a building of area 50 m^2 is represented by an area of 32 cm^2 .

Find the

(a) scale used to draw the map, in the form $1 : n$,

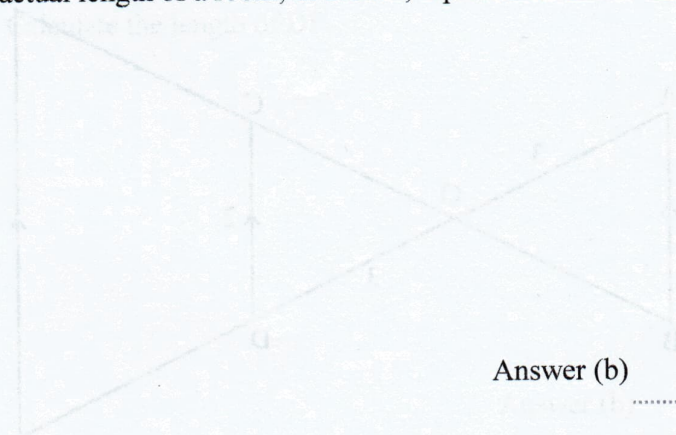
Answer (a)

[3]



15

- (b) actual length of a room, in metres, represented on the map by a line 6cm long .



Answer (b)

[2]

- 17 Given that $f(x) = x^2 + x - 20$
find

(a) $f(1)$,

Answer (a)

[2]

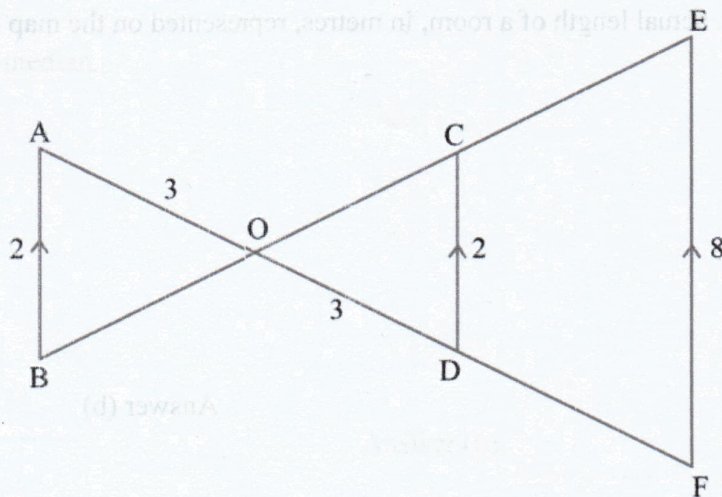
- (b) the values of x for which $f(x) = 0$.

Answer (b)

[3]



18



In the diagram, AB, CD and EF are parallel lines.

Lines AF and BE intersect at O.

$AB = CD = 2\text{cm}$ and $AO = OD = 3\text{cm}$.

$EF = 8\text{cm}$.

- (a) Name the triangle which is
- (i) congruent to triangle OAB ,

Answer (a)(i)

[1]

- (ii) similar to triangle OAB .

Answer (a)(i)

[1]



Candidate Name

Centre Number

Candidate Number

17

- (b) Calculate the length of DF.

Answer (b)

[3]

- 19 (a) Make x the subject of the formula

$$p = \frac{k}{\sqrt{x}}$$

Answer (a)

[3]

- (b) Solve the simultaneous equations:

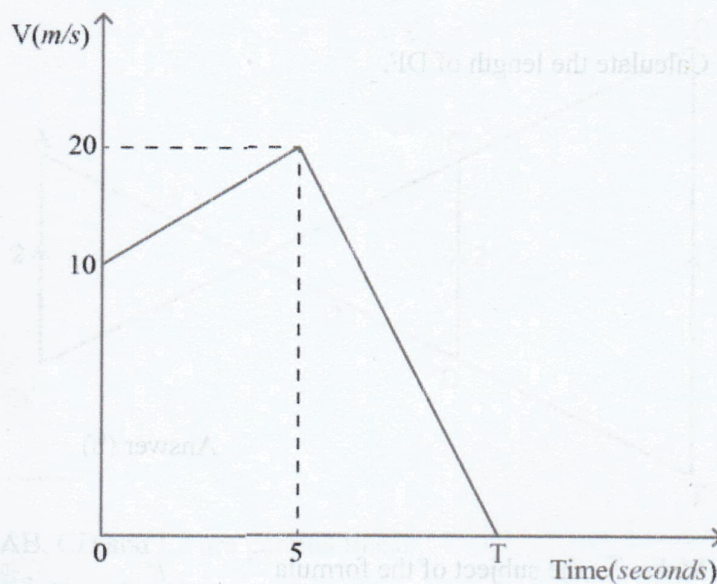
$$3x + 2y = 4$$

$$2y + x = 0$$

Answer (b)

[3]

20



The graph shows the movement of a particle which accelerates from 10m/s to 20m/s in 5 seconds. It then retards uniformly at 4m/s^2 until it comes to rest, at time, T , seconds. Use the graph to calculate,

- (a) the acceleration of the particle during the first 5 seconds,

Answer (a)

[2]

- (b) T , the total time taken for the journey,

Answer (b)

[2]



Candidate Name

Centre Number

Candidate Number

19

- (c) the distance travelled by the particle during the first 5 seconds.

Answer (c)

[2]

- 21 A quantity C is the sum of two parts. The first part varies directly as the cube of x and the second part varies inversely as the square of x .

- (a) Write down the equation connecting C and x using constants a and b .

Answer (a)

[1]

- (b) Given that $C = 74$ when $x = 1$ and $C = 34$ when $x = 2$, find the

- (i) value of a and the value of b ,

Answer (b)(i)

[3]

Candidate Name

Centre Number

Centre Number

Candidate Number

--	--	--	--

20

(ii) value of C when $x = 3$.

Answer (b)(ii)

[2]

22

(a) Evaluate $\begin{pmatrix} 1 & -2 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 1 & 1 \end{pmatrix}$.

Answer (a)

[2]

(b) Find the inverse of $\begin{pmatrix} 4 & 2 \\ 1 & 1 \end{pmatrix}$.

Answer (b)

[3]



Candidate Name

Centre Number

Candidate Number

--	--	--

21

- (c) Given that matrix $A = \begin{pmatrix} 1 & 8 \\ 2 & y^2 \end{pmatrix}$ and that it is a singular matrix,
find the possible values of y .

Answer (c)

[2]

